

SHREYA ROY

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EDUCATION

SRM Institute of Science & Technology

B.Tech. in Computer Science and Engineering

- GPA: 8.79/10; Finalist in multiple hackathons.
- Currently working on a research paper on Major Depressive Disorder.

Chennai, Tamil Nadu

2024 – 2028

Assembly of God Church School

PCMB

- ISC: 72.4% aggregate.

Asansol, West Bengal

2022 – 2024

Assembly of God Church School

Science

- ICSE: 87% overall; Distinction.

Asansol, West Bengal

2020 – 2022

TECHNICAL SKILLS

Programming Languages: Java, C++, Python, TypeScript, JavaScript, SQL

Back-End & Cloud: Node.js, Express, FastAPI, REST APIs, PostgreSQL, MongoDB, Firebase, Cloud Deployment

AI/ML & Analytics: Multi-Agent Systems, Machine Learning, Acoustic Event Classification, Neuro-Symbolic AI

Front-End & UI: React.js, Tailwind CSS, Responsive Design, Recharts, Figma

Developer Tools & Concepts: Git/GitHub, Data Structures & Algorithms, Object-Oriented Programming (OOP), System Architecture, Cross-Functional Collaboration

RESEARCH

Major Depressive Disorder (MDD)

- Investigating machine learning approaches for early detection of Major Depressive Disorder using EEG signals and identifying neural patterns associated with depression.
- Analyzing EEG signals to evaluate their potential as an objective complement to conventional diagnosis methods.

PROJECTS

AirDrawer [GitHub]

- Developed a browser-based AI spatial canvas that enables users to draw, manipulate, and export 3D models using hand gestures captured through a standard webcam.
- Implemented kinematic coordinate mapping and real-time matrix transformations to convert webcam input into a low-latency virtual 3D environment.

EchoMind [GitHub]

- Built an AI-powered acoustic surveillance platform that analyzes ambient audio in real time using acoustic event classification.
- Detects threats such as gunshots, breaking glass, and human screams, with explainable AI and automated Twilio SMS alerts.

Phishing Sentinel [GitHub]

- Developed a real-time phishing detection system using a hybrid machine learning pipeline, Go API gateway, FastAPI inference service, React dashboard, and Chrome extension.
- Implemented DOM and URL feature analysis with a Balanced Random Forest classifier trained on 79,760 websites, achieving 94.58% accuracy.

SafeRide Shield [GitHub]

- Developed an AI-driven parametric income insurance platform for gig workers using external data triggers to automate payouts without manual claims.
- Implemented a “Rain Curfew” mechanism that compensates riders when adverse weather conditions prevent them from working.

LANGUAGES

English: Professional **Hindi:** Professional **Bengali:** Native